Ericsson PE Node JAY-PE-09 Optical Transport Failure ALARM_3670 Resolution SOP

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_3670

Description

In the context of the Ericsson PE node JAY-PE-09, an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) manifests due to the degradation of the optical signals transmitted over the SFP (Small Form Factor Pluggable) or XFP (10G Small Form Factor Pluggable) transceivers. This failure domain is specific to the PE role on the Ericsson equipment as it pertains to the Optical Transport Network (OTN) layer, which is responsible for transporting client signals over optical fibers. The OTN layer relies on the SFP/XFP transceivers to convert electrical signals into optical signals, which are then transmitted over the fiber. A degradation in the SFP/XFP transceivers can lead to a loss of signal quality, resulting in the ALARM_3670 alarm code being triggered. Moreover, fiber cuts or Rx/Tx power loss can also contribute to this failure domain, further emphasizing the importance of regular maintenance and monitoring of the optical transport network.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Jayanagar Metro Site for physical access and assistance.
- Verify the status of the redundant link to ensure it is operational and can take over in case of a failure.
- Check the optical fiber connections for any signs of damage or degradation.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface optics, Expected Output: Optical interface information, including SFP/XFP transceiver status and fiber connection details, Description: This command provides detailed information about the optical interfaces on the PE node, including the status of the SFP/XFP transceivers and fiber connections. By analyzing the output, we can determine if there are any issues with the SFP/XFP transceivers or fiber connections that may be contributing to the ALARM_3670 alarm code.

Command: show alarms detail, Expected Output: Detailed alarm information, including the ALARM_3670 alarm code and any related alarms, Description: This command provides detailed information about the alarms on the PE node, including the ALARM_3670 alarm code and any
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related alarms. By analyzing the output, we can determine the cause of the ALARM_3670 alarm code and identify any related issues that may need to be addressed.
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Resolution Steps

Step 1: Step: Replace SFP/XFP transceiver, Command: replace optics, Description: Replace the SFP/XFP transceiver with a new one to resolve the issue. Ensure the new transceiver is properly installed and configured.

Step 2: Step: Repair or replace fiber, Command: repair fiber, Description: Repair or replace the fiber connection to resolve the issue. Ensure the fiber connection is properly installed and configured.

Post-Check Verification

```
Command: show interface optics, Expected Output: Optical interface information, including SFP/XFP transceiver status and fiber connection details, Description: Verify that the issue has been resolved by checking the status of the SFP/XFP transceivers and fiber connections using the show interface optics command.

Command: show alarms detail, Expected Output: Detailed alarm information, including the absence of the ALARM_3670 alarm code, Description: Verify that the ALARM_3670 alarm code has been cleared by checking the alarm information using the show alarms detail command.
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Roll-back Procedures

- Step: Restore original SFP/XFP transceiver, Command: restore optics, Description: Restore the original SFP/XFP transceiver if the replacement transceiver does not resolve the issue.

- Step: Repair or replace original fiber, Command: restore fiber, Description: Repair or replace the original fiber connection if the replacement fiber does not resolve the issue.